

Conformism–Individualism Trade-offs in LQG Graphon Mean Field Games with Control Mean Field Costs[★]

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Abstract: Limitations on the power or resources available to individual agents frequently arise in real-world games. To model such situations, this work studies a class of Linear Quadratic Gaussian Graphon Mean Field Games (LQG–GMFG) whose cost functional incorporates quadratic penalties on deviations from both an agent’s privately desired control and its local control mean field. These penalties represent two distinct drivers: individualism (acting on private preferences) and conformism (avoiding the higher resource costs incurred when acting differently from others). Separate state and control mean-field consistency conditions are imposed, and a sufficient condition for existence is established. Using spectral decomposition, an explicit value function is obtained for the infinite-horizon, exponentially discounted stationary case, and numerical simulations reveal a trade-off between conformism and individualism.

Keywords: Graphon Mean Field Games, Linear Quadratic Gaussian, Resource Constraints, Value Function Characterization, Networked Multi-Agent Systems

1. INTRODUCTION

In many real-world games, agents often face limitations on the resources or energy they can exploit. Moreover, the available resources of each agent may depend on the actions chosen by all other agents. In mathematical models, this interdependence is often treated by imposing shared feasibility constraints—where each agent’s available strategy set is directly restricted by the collective actions of all other agents—leading to the notion of Generalized Nash Equilibrium (GNE) (Facchinei and Kanzow, 2010). For dynamic games with shared constraints across all agents, the differential variational inequality (DVI) framework has been developed to characterize GNE (Chen and Wang, 2014).

The above scenarios concern games with a finite number of agents. As the number of agents grows large and tends to infinity, the resulting models are known as Mean Field Games (MFG), introduced in (Huang et al., 2006). Finding a Nash Equilibrium (NE) in an MFG amounts to identifying a consistent pair consisting of an optimal control and a probability density flow of states. Consistency requires that the optimal control solves the Hamilton–Jacobi–Bellman (HJB) equation parameterized by the density flow, while the flow itself is reproduced when each agent applies the optimal control.

The classical MFG framework was extended to graphon mean field games (GMFG) to allow heterogeneous cou-

plings between agents (Caines and Huang, 2021). A graphon can be viewed as the limit of a sequence of finite graphs, with the limit object defined in the cut metric space induced by the cut norm (Lovász, 2012).

To the best of our knowledge, the notion of GNE and shared constraints among agents has not been widely studied in the MFG or GMFG settings. A recent contribution in (Hintermüller et al., 2024) considered a GNE problem with mean field interactions under linear deterministic dynamics. At the same time, several works have examined MFGs with state, control, or joint state–control constraints. When both state and control evolutions are analyzed simultaneously, the resulting models are often termed Mean Field Games of Controls (MFGC) or extended MFGs, with a general analysis presented in (Cardaliaguet and Lehalle, 2018). For MFGC with additional inequality-type mixed constraints, (Bonnans et al., 2023) used a Lagrangian approach to establish existence and uniqueness of an auxiliary MFGC equilibrium consisting of a state trajectory and a corresponding Lagrange multiplier.

While such strictly coupled constraints necessitate equilibrium analysis within narrower, interdependent feasible sets, an alternative and practical approach is to model these limitations via cost penalization. In this paradigm, the control space remains unconstrained, but agents incur increasingly higher costs as they deviate from a preferred baseline or the collective behavior of the population. Applying this paradigm, the present work introduces control mean field penalties—acting as soft constraints—into

[★] This work is supported in part by AFOSR Grant OSR FA9550-23-1-0015 and NSERC (Canada) Grant RGPIN-2019-05336.

the LQG-GMFG framework. By having agents internalize these deviations costs, the formulation structurally captures the tension between individual preference and population-level conformity, a trade-off that we subsequently illustrate via numerical simulations. Specifically, the main contributions are:

- Develop an LQG-GMFG framework in which the state mean-field and control mean-field consistency conditions are imposed separately, enabling independent modeling of state and control mismatches in large populations.
- Derive the Forward-Backward ODE (FB-ODE) system that yields the optimal control when the cost includes a term associated with the control mean field, making the influence of the corresponding penalty matrix explicit.
- Obtain an explicit value function formula in the infinite-horizon, exponentially discounted stationary LQG setting in which the cost functional includes a control mean-field term.

These contributions provides a foundation for future studies of more challenging problems involving local hard control constraints, nonlinear dynamics and costs, and globally coupled constraints.

2. PRELIMINARIES AND NOTATION

Let $\alpha \in [0, 1]$ be a node index. For simplicity, assume that the state variable $x_\alpha(t)$ and $u_\alpha(t)$ are of the same dimension, n . Let $\mathbf{M} : [0, 1]^2 \rightarrow [-c, c]$ denote a graphon, i.e., a bounded symmetric Lebesgue measurable function. The space of all such functions is denoted by $\mathcal{W}_c$. Assume that at each time instant t , the state variables $x_\alpha(t)$ and the control $u_\alpha(t)$ across all node indexes are in the Hilbert space $(L^2[0, 1])^n$ with the norm $\|\mathbf{v}\|_{(L^2[0, 1])^n} = \left(\int_{[0, 1]} \|\mathbf{v}(\alpha)\|_{\mathbb{R}^n}^2 d\alpha \right)^{\frac{1}{2}}$. We use $L^2([0, T]; (L^2[0, 1])^n)$ to denote a Hilbert space of equivalence classes of strongly measurable (in the Böcher sense) mappings from $[0, T]$ to $(L^2[0, 1])^n$ that are integrable with the norm $\|\mathbf{x}\|_{L^2([0, T]; (L^2[0, 1])^n)} = \left(\int_0^T \|\mathbf{x}(t)\|_{(L^2[0, 1])^n}^2 dt \right)^{\frac{1}{2}}$.

Now we can define how a graphon operate on an element in $(L^2[0, 1])^n$. We can associate each graphon a linear bounded operator $\mathbb{T} \in \mathcal{L}([0, T]; (L^2[0, 1])^n)$. For a matrix $D \in \mathbb{R}^n$ and $\mathbf{M} \in \mathcal{W}_c$ the operator $[D\mathbf{M}]$ is defined as follows,

$$([D\mathbf{M}]\mathbf{v})(\alpha) \triangleq D \begin{pmatrix} (\mathbf{M}v_1)(\alpha) \\ \vdots \\ (\mathbf{M}v_n)(\alpha) \end{pmatrix} \in \mathbb{R}^n. \quad (1)$$

Consider the time-varying graphon dynamical system,

$$\dot{\mathbf{x}}(t) = [A(t)\mathbb{I} + D(t)\mathbf{M}]\mathbf{x}(t) + [B(t)\mathbb{I} + E(t)\mathbf{M}]\mathbf{u}(t) \quad (2)$$

where the matrix functions are assumed to be continuous. Let $\mathbb{A}(t) = A(t)\mathbb{I} + D(t)\mathbf{M}$ and $\mathbb{B}(t) = B(t)\mathbb{I} + E(t)\mathbf{M}$. A mild solution to (2) is continuous over $[0, T]$ and satisfies

$$\mathbf{x}(t) = \mathbf{x}_0 + \int_0^t \mathbb{A}(\tau)\mathbf{x}(\tau) + \mathbb{B}(\tau)\mathbf{u}(\tau) d\tau. \quad (3)$$

Consider the initial value problem,

$$\dot{\mathbf{x}}(t) = \mathbb{A}(t)\mathbf{x}(t), \quad \mathbf{x}(s) = \mathbf{x}_s \quad (4)$$

the classical to the problem exists by the assumption.

Definition 1. (Evolution operator) The evolution operator associated with $\mathbb{A}$ is defined as the two-parameter family of operators $\Phi(\cdot, \cdot)$ satisfy $\Phi(t, s)\mathbf{x}_s = \mathbf{x}_t$, where $\mathbf{x}(t)$ is the classical solution to (4).

By (Gao et al., 2023, Lem.1), the unique mild solution to (2) is given by

$$\mathbf{x}(t) = \Phi(t, 0)\mathbf{x}_0 + \int_0^t \Phi(t, \tau)[B(\tau)\mathbb{I} + E(\tau)\mathbf{M}]\mathbf{u}(\tau) d\tau. \quad (5)$$

If $\mathbf{u}$ is continuous in $[0, T]$ in addition, the solution given by (5) is also a classical solution by (Gao et al., 2023).

3. LQG-GMFG EQUATIONS WITH CONTROL MEAN FIELD COST

This section presents the solution of an LQG-GMFG problem that includes a control mean-field cost. The formulation incorporates desired control trajectories and time-varying penalty matrices associated with deviations of the private control from its desired control and from the local control mean field.

3.1 A LQG-GMFG Problem with Control Mean Field Cost

Let the dynamics of a generic agent located at node α be given by,

$$dx_\alpha(t) = (Ax_\alpha(t) + Bu_\alpha(t) + D_0\bar{x}_\alpha(t) + Dz_\alpha(t))dt + \Sigma dw_\alpha(t), \quad (6)$$

where $z_\alpha(t) := \int_0^1 \mathbf{M}(\alpha, \beta)\bar{x}_\beta(t)d\beta$ denotes the local state network mean field, and $\mathbf{M}$ is the underlying graphon.

Let the cost function of agents at node α be given by,

$$\begin{aligned} J_\alpha(x_\alpha(t), u_\alpha(t); \bar{x}_\alpha(t), z_\alpha(t), \bar{u}_\alpha(t)) \\ = \mathbb{E} \int_0^T \|x_\alpha(t) - \nu_\alpha(t)\|_Q^2 + \|u_\alpha(t) - \xi_\alpha(t)\|_{R_1(t)}^2 \\ + \|u_\alpha(t) - \bar{u}_\alpha(t)\|_{R_2(t)}^2 dt + \mathbb{E} \|x_\alpha(T) - \nu_\alpha(T)\|_{Q_T}^2, \end{aligned} \quad (7)$$

where $\nu_\alpha(t) := H_0\bar{x}_\alpha(t) + Hz_\alpha(t) + \eta$ is the signal that each agent wants to track with their state variable. $\xi_\alpha(t)$ is the control that each agent wants to track with. $\bar{u}_\alpha(t)$ is the local control mean field.

By dynamic programming principle, the value function $V_\alpha(t, x_\alpha(t); \bar{x}_\alpha(t), z_\alpha(t), \bar{u}_\alpha(t))$ solves the following HJB equation, (Yong and Zhou, 1999, Chap.4, Prop.3.5),

$$\begin{aligned} - \frac{\partial V_\alpha(t, x_\alpha(t); \bar{x}_\alpha(t), z_\alpha(t), \bar{u}_\alpha(t))}{\partial t} \\ = \inf_{u_\alpha \in U} \left\{ \|x_\alpha(t) - \nu_\alpha(t)\|_Q^2 + \|u_\alpha - \xi_\alpha(t)\|_{R_1(t)}^2 \right. \\ \left. + \|u_\alpha - \bar{u}_\alpha(t)\|_{R_2(t)}^2 \right. \\ \left. + (Ax_\alpha(t) + Bu_\alpha + D_0\bar{x}_\alpha(t) + Dz_\alpha(t))^T \right. \\ \left. \frac{\partial V_\alpha(t, x_\alpha(t); \bar{x}_\alpha(t), z_\alpha(t), \bar{u}_\alpha(t))}{\partial x_\alpha(t)} \right\} \\ + \text{tr} \left\{ \Sigma \Sigma^T \frac{\partial^2 V_\alpha(t, x_\alpha(t); \bar{x}_\alpha(t), z_\alpha(t), \bar{u}_\alpha(t))}{\partial x_\alpha(t)^2} \right\}. \end{aligned} \quad (8)$$

We use the Ansatz,

$$V_\alpha(t, x_\alpha(t); \bar{x}_\alpha(t), z_\alpha(t); \bar{u}_\alpha(t)) = x_\alpha(t)^T \Pi_\alpha(t) x_\alpha(t) + 2s_\alpha(t)^T x_\alpha(t) + r_\alpha(t). \quad (9)$$

By the stochastic verification theorem, (Yong and Zhou, 1999, Chap.5, Thm.4.1), a control $u_\alpha(t), t \in [0, T]$ is optimal if and only if it minimizes the Hamiltonian. Thus, the optimal control is given by,

$$u_\alpha^o(t) = [R_1(t) + R_2(t)]^{-1} \{R_1(t)\xi_\alpha(t) + R_2(t)\bar{u}_\alpha(t) - B^T[\Pi_\alpha(t)x_\alpha(t) + s_\alpha(t)]\}. \quad (10)$$

Control Mean Field Consistency Condition

By taking expectation on both sides of (10) and combine terms with $\bar{u}_\alpha(t)$, we get the consistency condition,

$$\bar{u}_\alpha(t) = \xi_\alpha(t) - R_1(t)^{-1} B^T[\Pi_\alpha(t)\bar{x}_\alpha(t) + s_\alpha(t)]. \quad (11)$$

Substitute (11) back into (10), we can eliminate the term with $\bar{u}_\alpha(t)$ in the optimal control expression,

$$u_\alpha^o(t) = \xi_\alpha(t) - [R_1(t) + R_2(t)]^{-1} B^T \Pi_\alpha(t) x_\alpha(t) - [R_1(t) + R_2(t)]^{-1} R_2(t) R_1(t)^{-1} B^T \Pi_\alpha(t) \bar{x}_\alpha(t) - R_1(t)^{-1} B^T s_\alpha(t). \quad (12)$$

Substituting (12) back to the HJB equation (8), expanding all terms, and equating coefficients of terms with the same order of $x_\alpha(t)$, we get ODEs for $\Pi_\alpha(t)$ and $s_\alpha(t)$,

$$\begin{aligned} -\dot{\Pi}_\alpha(t) &= Q + \Pi_\alpha(t)A + A^T \Pi_\alpha(t) \\ &\quad - \Pi_\alpha(t)B[R_1(t) + R_2(t)]^{-1} B^T \Pi_\alpha(t), \\ \Pi_\alpha(T) &= Q_T. \end{aligned} \quad (13)$$

$$\begin{aligned} \dot{s}_\alpha(t) &= -[A - BR_1(t)B^T \Pi_\alpha(t)]^T s_\alpha(t) \\ &\quad + \{\Pi_\alpha(t)B[R_1(t) + R_2(t)]^{-1} R_2(t)R_1(t)^{-1} B^T \Pi_\alpha(t) \\ &\quad + QH_0 - \Pi_\alpha(t)D\} \bar{x}_\alpha(t) - \Pi_\alpha(t)B\xi_\alpha(t) \\ &\quad + [QH - \Pi_\alpha(t)D]z_\alpha(t) + Q\eta, \\ s_\alpha(T) &= -Q_T \nu_\alpha(T). \end{aligned} \quad (14)$$

Since ODEs of $\Pi_\alpha(t)$ are the same for all different $\alpha \in [0, 1]$, we let $\Pi(t) := \Pi_\alpha(t), \alpha \in [0, 1]$.

State Mean Field Consistency Condition

By taking expectation on both sides of the dynamics of the state (6), and substituting (11) into it, we get the ODE for the state mean field,

$$\begin{aligned} \dot{\bar{x}}_\alpha(t) &= [A - BR_1(t)B^T \Pi(t) + D_0]\bar{x}_\alpha(t) \\ &\quad - BR_1(t)^{-1} B^T s_\alpha(t) + B\xi_\alpha(t) + Dz_\alpha(t), \\ \bar{x}_\alpha(0) &= m_\alpha. \end{aligned} \quad (15)$$

3.2 A Sufficient Condition for the Existence of the Global FB-ODE System

Assume that a graphon $\mathbb{W}(\alpha, \beta)$ is in the space $L^2([0, 1]^2)$, then its associated operator is compact. (Conway, 2019, Chap.2, Prop.4.7) Moreover, a graphon operator is a self-adjoint operator, thus it can be expressed as a countably infinite sum of its eigenvalues times its orthogonal projection operators onto the corresponding eigenspaces. (Conway, 2019, Chap.4, Thm.5.1)

Thus, we can write the global forward-backward equations,

$$\begin{aligned} \dot{\bar{x}}(t) &= [(A - BR_1(t)B^T \Pi(t) + D_0)\mathbb{I}]\bar{x}(t) \\ &\quad - [BR_1(t)^{-1} B^T \mathbb{I}]s(t) + [B\mathbb{I}]\xi(t) + [D\mathbb{M}]\bar{x}(t), \\ \bar{x}(0) &= m. \end{aligned} \quad (16)$$

$$\begin{aligned} \dot{s}(t) &= -[(A - BR_1(t)B^T \Pi(t))^T \mathbb{I}]s(t) \\ &\quad + [(\Pi(t)B(R_1(t) + R_2(t))^{-1} R_2(t)R_1(t)^{-1} B^T \Pi(t) \\ &\quad + QH_0 - \Pi(t)D)\mathbb{I}]\bar{x}(t) - [\Pi(t)B\mathbb{I}]\xi(t) \\ &\quad + [(QH - \Pi(t)D)\mathbb{M}]\bar{x}(t) + [Q\mathbb{I}](\eta\mathbf{1}), \\ s(T) &= -[Q_T H_0 \mathbb{I} + Q_T H \mathbb{M}]\bar{x}(T) - [Q_T \mathbb{I}](\eta\mathbf{1}). \end{aligned} \quad (17)$$

We have a proposition that gives us condition for a unique solution. Proof for this proposition applied a contraction principle in the space $C([0, T]; (L^2[0, 1])^n)$.

Let the evolution operator of $[(A - BR_1(t)B^T \Pi(t) + D_0)\mathbb{I} + D\mathbb{M}]$ and $[-(A - BR_1(t)B^T \Pi(t))^T \mathbb{I}]$ be $\phi_1^{\mathbf{M}}(\cdot, \cdot)$, and $\phi_2(\cdot, \cdot)$ respectively. Let $\|\cdot\|_{op}$ denote the operator norm for $\mathcal{L}((L^2[0, 1])^n)$. And we define a quantity as follows,

$$\begin{aligned} L_0(\mathbf{M}) &= \sup_{t \in [0, T]} \left\{ \int_0^t \|\phi_1^{\mathbf{M}}(t, \tau)[BR_1(\tau)^{-1} B^T \mathbb{I}]\phi_2(\tau, T) \right. \\ &\quad \left. [Q_T H_0 \mathbb{I} + Q_T H \mathbb{M}]\|_{op} d\tau \right\} + \sup_{t \in [0, T]} \left\{ \int_0^t \int_r^T \|\phi_1^{\mathbf{M}}(t, \tau) \right. \\ &\quad \left. [BR_1(\tau)^{-1} B^T \mathbb{I}]\phi_2(\tau, r) \left[(\Pi(r)B(R_1(r) + R_2(r))^{-1} \right. \right. \right. \\ &\quad \left. \left. R_2(r)R_1(r)^{-1} B^T \Pi(r) + QH_0 - \Pi(r)D_0)\mathbb{I} \right. \right. \\ &\quad \left. \left. + (QH - \Pi(r)D)\mathbb{M} \right] \right\|_{op} dr d\tau \right\}. \end{aligned} \quad (18)$$

Proposition 2. The LQG-GMFG global equations (17) and (16) has a unique classical solution if $L_0(\mathbf{M}) < 1$.

Proof. The classical solutions to the FB-ODE systems are given by the following convolution formula,

$$\begin{aligned} \bar{x}(t) &= \phi_1^{\mathbf{M}}(t, 0)m + \int_0^t \phi_1^{\mathbf{M}}(t, \tau) \left([(-BR_1(\tau)^{-1} B^T \mathbb{I})] \right. \\ &\quad \left. s(\tau) + [B\mathbb{I}]\xi(\tau) \right) d\tau. \end{aligned} \quad (19)$$

$$\begin{aligned} s(\tau) &= \phi_2(\tau, T)s(T) \\ &\quad - \int_r^T \phi_2(\tau, r) \left([(\Pi(r)B(R_1(r) + R_2(r))^{-1} R_2(r)R_1(r)^{-1} \right. \\ &\quad \left. B^T \Pi(r) + QH_0 - \Pi(r)D)\mathbb{I} + (QH - \Pi(r)D)\mathbb{M}]\bar{x}(r) \right. \\ &\quad \left. - [\Pi(r)B\mathbb{I}]\xi(r) + [Q\mathbb{I}](\eta\mathbf{1}) \right) dr. \end{aligned} \quad (20)$$

Substituting $s(\tau)$ in (20) to (19) yields,

$$\begin{aligned}
\bar{\mathbf{x}}(t) &= \phi_1^{\mathbf{M}}(t, 0)\mathbf{m} - \int_0^t \phi_1^{\mathbf{M}}(t, \tau) [(BR_1(\tau)^{-1}B^T)\mathbb{I}] \\
&\left(\phi_2(\tau, T) (-[Q_T H_0 \mathbb{I} + Q_T H \mathbf{M}]\bar{\mathbf{x}}(T) - [Q_T \mathbb{I}](\eta \mathbf{1})) \right. \\
&- \int_r^T \phi_2(\tau, r) \left([(\Pi(r)B(R_1(r) + R_2(r))^{-1}R_2(r)R_1(r)^{-1} \right. \\
&B^T \Pi(r) + QH_0 - \Pi(r)D)\mathbb{I} + (QH - \Pi(r)D)\mathbf{M}]\bar{\mathbf{x}}(r) \\
&- [\Pi(r)B\mathbb{I}]\boldsymbol{\xi}(r) + [Q\mathbb{I}](\eta \mathbf{1})) dr \Big) d\tau \\
&+ \int_0^t \phi_1^{\mathbf{M}}(t, \tau) [B\mathbb{I}]\boldsymbol{\xi}(\tau) d\tau. \tag{21}
\end{aligned}$$

Neglecting all terms in (21) independent of $\bar{\mathbf{x}}(t)$, a standard contraction argument yields the result. $\square$

3.3 Spectral Decomposition of the FB-ODE System

Assumption 3. The eigenfunctions $\{\mathbf{f}_\ell\}_{\ell \in \mathcal{I}_\lambda}$ form an orthonormal basis in $L^2[0, 1]$, i.e., it is complete.

Remark 4. This completeness assumption ensures that the eigenfunctions span $L^2[0, 1]$ (no nontrivial complement). The UAG graph meets this requirement.

Remark 5. Since $L^2[0, 1]$ is a separable Hilbert space, we can index $\{\mathbf{f}_\ell\}_{\ell \in \mathcal{I}_\lambda}$ by ℓ from 1 to ∞ .

Next, if we have the eigenvalues λ_ℓ and the eigenfunctions $\mathbf{f}_\ell$ of the operators appear in the above equations, (actually, these operators share same eigenvalues and eigenfunctions (Gao and Caines, 2021)) we can transform the equations into following simpler equations,

$$\begin{aligned}
\dot{\bar{x}}_\ell(t) &= (A - BR_1(t)^{-1}B^T \Pi(t) + D_0)\bar{x}_\ell(t) \\
&- BR_1(t)^{-1}B^T s_\ell(t) + B\xi_\ell(t) + \lambda_\ell D\bar{x}_\ell(t), \\
\bar{x}_\ell(0) &= \langle \mathbf{m}, \mathbf{f}_\ell \rangle. \\
\dot{s}_\ell(t) &= - (A - BR_1(t)^{-1}B^T \Pi(t))^T s_\ell(t) \\
&+ \left(\Pi(t)B(R_1(t) + R_2(t))^{-1}R_2(t)R_1(t)^{-1}B^T \Pi(t) \right. \\
&+ QH_0 - \Pi(t)D \Big) \bar{x}_\ell(t) + \lambda_\ell (QH - \Pi(t)D)\bar{x}_\ell(t) \\
&- \Pi(t)B\xi_\ell(t) + Q\eta \langle \mathbf{1}, \mathbf{f}_\ell \rangle. \\
s_\ell(T) &= -Q_T H_0 \bar{x}_\ell(T) - \lambda_\ell Q_T H \bar{x}_\ell(T) - Q_T \eta \langle \mathbf{1}, \mathbf{f}_\ell \rangle. \tag{22}
\end{aligned}$$

After writing their decomposed equations, we have the following proposition.

Proposition 6. If the global forward-backward ODE (17) (16) has a unique classical solution, then it satisfies (22) and (23).

4. VALUE FUNCTION IN INFINITE HORIZON EXPONENTIALLY DISCOUNTED COST CASE

In order to obtain a formula for the value function illustrating the comparative effect of $R_1(t)$ and $R_2(t)$, we employ a similar setting and approach in (Tchuendom et al. (2022)).

Assume all the variable in the model are scalars. Let $A = 0, D_0 = 0, D = 0, H_0 = 0, H = 1, \eta = 0, Q = 1$, and corresponding uppercase letters are changed into lowercase letters. Specifically, the model reduces to

$$dx_\alpha(t) = bu_\alpha(t) + \sigma dw_\alpha(t), \quad x_\alpha(0) \sim \mathcal{N}(m_\alpha, \nu^2) \tag{24}$$

$$\begin{aligned}
J_\alpha(t, x_\alpha(t); z_\alpha(t), \bar{u}_\alpha(t)) &= \mathbb{E} \int_0^\infty e^{-\rho t} \left((x_\alpha(t) - z_\alpha(t))^2 \right. \\
&+ r_1(u_\alpha(t))^2 + r_2(u_\alpha(t) - \bar{u}_\alpha(t))^2 \Big) dt. \tag{25}
\end{aligned}$$

This special case enables an analytical characterization of how the relative magnitudes of r_1 and r_2 shape the equilibrium behavior through their influence on the value function.

Because we are considering infinite horizon cost, the coefficient of the quadratic term in Ansatz of the value function is a constant,

$$V_\alpha(t, x_\alpha(t)) = \Gamma x_\alpha(t)^2 + 2s_\alpha(t)x_\alpha(t) + q_\alpha(t). \tag{26}$$

The value of π is determined by an algebraic Riccati equation,

$$\frac{1}{r_1 + r_2} b^2 \Gamma^2 + \rho \Gamma = 1. \tag{27}$$

And its value is taken to be the positive one,

$$\Gamma = -\frac{\rho(r_1 + r_2)}{2b^2} + \sqrt{\frac{\rho^2(r_1 + r_2)^2}{4b^4} + \frac{r_1 + r_2}{b^2}}. \tag{28}$$

Similarly, by applying verification theorem in stochastic optimal control, we get the following proposition stating how to obtain optimal control.

Proposition 7. To obtain the optimal control law for the problem determined by (24) and (25), one needs to solve the following ODE system in $L^2[0, 1]$,

$$\begin{aligned}
\dot{\bar{x}}_\alpha(t) &= -\frac{1}{r_1} b^2 \Gamma \bar{x}_\alpha(t) - \frac{1}{r_1} b^2 s_\alpha(t), \\
\bar{x}_\alpha(0) &= m_\alpha, \quad t \in [0, \infty), \\
\dot{s}_\alpha(t) &= \frac{1}{r_1} b^2 \Gamma s_\alpha(t) + b^2 \Gamma^2 \frac{r_2}{r_1(r_1 + r_2)} \bar{x}_\alpha(t) + z_\alpha(t) \\
&+ \rho s_\alpha(t), \quad t \in [0, \infty). \tag{29}
\end{aligned}$$

Then, the optimal control is given by

$$\begin{aligned}
u_\alpha^o(t) &= -\frac{1}{r_1 + r_2} b \pi x_\alpha(t) - \frac{r_2}{r_1(r_1 + r_2)} b \pi \bar{x}_\alpha(t) \\
&- \frac{1}{r_1} b s_\alpha(t), \quad t \in [0, \infty). \tag{30}
\end{aligned}$$

These formula are particularly degenerated forms in proposition 6. We also obtain an ODE for $q_\alpha(t)$ when we applying verification theorem to the problem, it is stated in the following proposition.

Proposition 8. The process $q_\alpha(t)$ satisfies the following ODE,

$$\begin{aligned}
\dot{q}_\alpha(t) &= -\sigma^2 \Gamma + \frac{1}{r_1} b^2 s_\alpha(t) - \frac{r_2}{r_1(r_1 + r_2)} b^2 \Gamma^2 \bar{x}_\alpha(t)^2 \\
&+ \rho q_\alpha(t) - z_\alpha(t)^2, \quad t \in [0, \infty). \tag{31}
\end{aligned}$$

And the expectation of the value function evaluated at the initial time 0 is given by the following formula,

$$\bar{V}_\alpha(0) = \Gamma(\nu^2 + m_\alpha^2) + 2s_\alpha(0)m_\alpha + q_\alpha(0). \tag{32}$$

A technique for solving the system consists of (29), (30), and (32) is to assume that we are in the stationary state, where the processes are $\bar{x}_\alpha^\infty, s_\alpha^\infty, q_\alpha^\infty$ and satisfy the following equations,

$$0 = -\frac{1}{r_1}b^2\Gamma\bar{x}_\alpha^\infty - \frac{1}{r_1}b^2s_\alpha^\infty, \quad (34)$$

$$0 = \frac{1}{r_1}b^2\Gamma s_\alpha^\infty + b^2\Gamma^2 \frac{r_2}{r_1(r_1+r_2)}\bar{x}_\alpha^\infty + z_\alpha^\infty + \rho s_\alpha^\infty, \quad (35)$$

$$0 = -\sigma^2\Gamma + \frac{1}{r_1}b^2s_\alpha^\infty - \frac{r_2}{r_1(r_1+r_2)}b^2\Gamma^2\bar{x}_\alpha^\infty(t)^2 + \rho q_\alpha^\infty - (z_\alpha^\infty)^2. \quad (36)$$

By the first two equations, we obtain

$$\left[\left(\frac{b^2\Gamma^2}{r_1+r_2} + \rho\Gamma \right) \mathbb{I} - \mathbb{M} \right] \bar{x}_\alpha^\infty = 0, \quad (37)$$

and by the relation in the algebraic Riccati equation, we have $x_\alpha^\infty = 0$. And we have $s_\alpha^\infty = 0$ $q_\alpha^\infty = \frac{\sigma^2\Gamma}{\rho}$.

If we write the above variables into their spectral decomposition, we obtain the following proposition giving the coefficients of each component.

Proposition 9. We have the following spectral decomposition of $\bar{x}_\alpha(t)$ and $s_\alpha(t)$,

$$\bar{x}_\alpha(t) = \sum_{\ell=1}^{\infty} \mathbf{f}_\ell(\alpha) \bar{x}_\ell(t), \quad s_\alpha(t) = \sum_{\ell=1}^{\infty} \bar{\lambda}_\ell \mathbf{f}_\ell(\alpha) \bar{x}_\ell(t), \quad (38)$$

where for $\ell \in \{1, 2, \dots\}$, $\bar{\lambda}_\ell$ and $\bar{x}_\ell(t)$ are given by

$$\bar{\lambda}_\ell = \frac{\frac{\rho^2(r_1+r_2)r_2}{2r_1b^2} - \frac{\rho(r_1+r_2)r_2}{r_1b^2}\tilde{\theta}(0) + \frac{r_2}{r_1} + \lambda_\ell}{\frac{\rho r_2}{2r_1} - \theta(\lambda_\ell) - \frac{r_1+r_2}{r_1}\tilde{\theta}(0)}, \quad (39)$$

$$\bar{x}_\ell(t) = \langle \mathbf{m}, \mathbf{f}_\ell \rangle \exp \left[\left(\frac{\rho}{2} - \theta(\lambda_\ell) \right) t \right]. \quad (40)$$

And the functions $\theta, \tilde{\theta}$ are given by

$$\theta(\tau) = \sqrt{\frac{\rho^2}{4} + \frac{b^2}{r_1}(1-\tau)}, \quad (41)$$

$$\tilde{\theta}(\tau) = \sqrt{\frac{\rho^2}{4} + \frac{b^2}{r_1+r_2}(1-\tau)}. \quad (42)$$

Proof. In the infinite horizon exponentially discounted cost function case, for each index $\ell \in \{1, 2, \dots\}$, we have similar ODEs for $\bar{x}_\ell(t)$ and $s_\ell(t)$ as they are in Proposition 6,

$$\frac{d\bar{x}_\ell(t)}{dt} = -\frac{1}{r_1}b^2\Gamma\bar{x}_\ell(t) - \frac{1}{r_1}b^2s_\ell(t), \quad t \in [0, \infty)$$

$$\bar{x}_\ell(0) = \langle \mathbf{m}, \mathbf{f}_\ell \rangle, \quad (43)$$

$$\frac{ds_\ell(t)}{dt} = \frac{1}{r_1}b^2\Gamma s_\ell(t) + b^2\Gamma^2 \frac{r_2}{r_1(r_1+r_2)}\bar{x}_\ell(t) + \lambda_\ell \bar{x}_\ell(t) + \rho s_\ell(t), \quad t \in [0, \infty), \quad s_\ell^\infty = 0. \quad (44)$$

By variation of constants formula, and the relation in the algebraic Riccati equation (27) we have,

$$s_\ell(t) = -\int_t^\infty \exp \left(\left(\frac{b^2\Gamma}{r_1(r_1+r_2)} + \frac{1}{\Gamma} \right) (t-s) \right) \left(b^2\Gamma^2 \frac{r_2}{r_1(r_1+r_2)} + \lambda_\ell \right) \bar{x}_\ell(s) ds, \quad t \in [0, \infty). \quad (45)$$

Take derivative with respect to t on both sides of (19), substitute (45) into it, and using the relation in (27) we obtain,

$$\frac{d^2\bar{x}_\ell(t)}{dt^2} - \rho \frac{d\bar{x}_\ell(t)}{dt} + \frac{b^2}{r_1}(\lambda_\ell - 1)\bar{x}_\ell(t) = 0. \quad (46)$$

The characteristic equation of the ODE is given by,

$$\mu_\ell^2 - \rho\mu_\ell + \frac{b^2}{r_1}(\mu_\ell - 1) = 0. \quad (47)$$

Take the negative solution,

$$\mu_\ell = \frac{\rho}{2} - \sqrt{\frac{\rho^2}{4} + \frac{b^2}{r_1}(1-\lambda_{ell})} = \frac{\rho}{2} - \theta(\lambda_\ell) < 0. \quad (48)$$

Thus we have the trajectories for $\bar{x}_\ell(t)$, $\ell \in \{1, 2, \dots\}$,

$$\bar{x}_\ell(t) = \langle \mathbf{m}, \mathbf{f}_\ell \rangle \exp \left(\left(\frac{\rho}{2} - \theta(\lambda_\ell) \right) t \right), \quad t \in [0, \infty). \quad (49)$$

Substitute $\bar{x}_\ell(t)$ trajectories (49) into (20), and using the relation in (27), we obtain

$$s_\ell(t) = -\langle \mathbf{m}, \mathbf{f}_\ell \rangle \exp \left(\left(\frac{1}{r_1}b^2\Gamma + \rho \right) t \right) \left(b^2\Gamma^2 \frac{r_2}{r_1(r_1+r_2)} + \lambda_\ell \right) \int_t^\infty \exp \left(\left(\mu_\ell - \frac{b^2}{r_1}\Gamma - \rho \right) s \right) ds$$

$$= -\langle \mathbf{m}, \mathbf{f}_\ell \rangle \left(b^2\Gamma^2 \frac{r_2}{r_1(r_1+r_2)} + \lambda_\ell \right) \left(\mu_\ell - b^2\Gamma \frac{r_2}{r_1(r_1+r_2)} - \frac{1}{\Gamma} \right)^{-1} \exp(\mu_\ell t)$$

$$= -\langle \mathbf{m}, \mathbf{f}_\ell \rangle \frac{\frac{\rho^2(r_1+r_2)r_2}{2r_1b^2} - \frac{\rho(r_1+r_2)r_2}{r_1b^2}\tilde{\theta}(0) + \frac{r_2}{r_1} + \lambda_\ell}{\frac{\rho r_2}{2r_1} - \theta(\lambda_\ell) - \frac{r_1+r_2}{r_1}\tilde{\theta}(0)} \exp(\mu_\ell t), \quad t \in [0, \infty), \quad (50)$$

which completes the derivation. $\square$

Remark 10. Since we are working in the space $L^2[0, 1]$, the appearing countably infinite sum converge at $\alpha \in [0, 1]$, a.e.

Remark 11. If we have the following relation, $r_1 + r_2 = 1$, $\delta := r_2/r_1$, then the coefficients of $s_\ell(t)$ will have a cleaner form,

$$\bar{\lambda}_\ell(\delta) := -\frac{\delta b^2\Gamma^2 + \lambda_\ell}{\frac{\rho}{2} + (1+\delta)b^2\Gamma + \sqrt{\frac{\rho^2}{4} + b^2(1+\delta)(1-\lambda_\ell)}}. \quad (51)$$

The following proposition gives the explicit formula for the value function.

Proposition 12. We have the following expression for the expectation of the value function at the initial time 0,

$$\begin{aligned}
\bar{V}_\alpha(0) &= \Gamma\nu^2 + \Gamma(m_\alpha)^2 + 2m_\alpha \sum_{\ell=1}^{\infty} \bar{\lambda}_\ell \langle m, \mathbf{f}_\ell \rangle \mathbf{f}_\ell(\alpha) + \frac{\sigma^2 \Gamma}{\rho} \\
&+ \frac{1}{\rho} \left(\sum_{\ell=1}^{\infty} \lambda_\ell \langle m, \mathbf{f}_\ell \rangle \mathbf{f}_\ell(\alpha) \right)^2 - \frac{b^2}{r_1 \rho} \left(\sum_{\ell=1}^{\infty} \bar{\lambda}_\ell \langle m, \mathbf{f}_\ell \rangle \mathbf{f}_\ell(\alpha) \right)^2 \\
&+ \frac{r_2}{\rho r_1 (r_1 + r_2)} b^2 \Gamma^2 \left(\sum_{\ell=1}^{\infty} \langle m, \mathbf{f}_\ell \rangle \mathbf{f}_\ell(\alpha) \right)^2 \\
&- \sum_{k=1}^{\infty} \sum_{\ell=1}^{\infty} \left(\frac{\rho}{2} - \theta(\lambda_k) \right) (\theta(\lambda_k) + \theta(\lambda_\ell))^{-1} \langle m, \mathbf{f}_k \rangle \langle m, \mathbf{f}_\ell \rangle \\
&\mathbf{f}_k(\alpha) \mathbf{f}_\ell(\alpha) \left[\frac{2}{\rho} \lambda_k \lambda_\ell - \frac{2b^2}{\rho r_1} \bar{\lambda}_k \bar{\lambda}_\ell + \frac{2r_2}{\rho r_1 (r_1 + r_2)} b^2 \Gamma^2 \right]. \quad (52)
\end{aligned}$$

The expression (52) can be further simplified as follows.

Proposition 13. We have the following simplified expression for the value function,

$$\begin{aligned}
\bar{V}_\alpha(0) &= \Gamma\nu^2 + \Gamma(m_\alpha)^2 + 2m_\alpha \sum_{\ell=1}^{\infty} \bar{\lambda}_\ell \langle m, \mathbf{f}_\ell \rangle \mathbf{f}_\ell(\alpha) + \frac{\sigma^2 \Gamma}{\rho} \\
&- \sum_{k=1}^{\infty} \sum_{\ell=1}^{\infty} \left(\frac{\rho}{\theta(\lambda_k) + \theta(\lambda_\ell)} - 2 \right) \langle m, \mathbf{f}_k \rangle \langle m, \mathbf{f}_\ell \rangle \mathbf{f}_k(\alpha) \mathbf{f}_\ell(\alpha) \\
&\left[\frac{1}{\rho} \lambda_k \lambda_\ell - \frac{b^2}{\rho r_1} \bar{\lambda}_k \bar{\lambda}_\ell + \frac{r_2}{\rho r_1 (r_1 + r_2)} b^2 \Gamma^2 \right]. \quad (53)
\end{aligned}$$

To proceed the analysis, we assume that the initial distributions of the state is uniform in α .

Assumption 14. Assume that $m_\alpha = m, \forall \alpha \in [0, 1]$.

By the above assumption, we have the following more simpler formula for the value function.

Proposition 15. Under the assumptions 3 and 14, we have the following expression for the value function,

$$\begin{aligned}
\bar{V}_\alpha(0) &= \Gamma \left(\nu^2 + m^2 + \frac{\sigma^2}{\rho} \right) + 2m^2 \sum_{\ell=1}^{\infty} \bar{\lambda}_\ell \langle \mathbf{1}, \mathbf{f}_\ell \rangle \mathbf{f}_\ell(\alpha) \\
&- m^2 \sum_{k=1}^{\infty} \tilde{\lambda}_k^\alpha \langle \mathbf{1}, \mathbf{f}_k \rangle \mathbf{f}_k(\alpha), \quad (54)
\end{aligned}$$

where $\tilde{\lambda}_k^\alpha$ are given by

$$\begin{aligned}
\tilde{\lambda}_k^\alpha &:= \sum_{\ell=1}^{\infty} \langle \mathbf{1}, \mathbf{f}_\ell \rangle \mathbf{f}_\ell(\alpha) \left(\frac{\rho}{\theta(\lambda_k) + \theta(\lambda_\ell)} - 2 \right) \\
&\left[\frac{1}{\rho} \lambda_k \lambda_\ell - \frac{b^2}{\rho r_1} \bar{\lambda}_k \bar{\lambda}_\ell + \frac{r_2}{\rho r_1 (r_1 + r_2)} b^2 \Gamma^2 \right]. \quad (55)
\end{aligned}$$

To analyze the effect of δ , we make the following assumption.

Assumption 16. Assume that $\langle \mathbf{1}, \mathbf{f}_1 \rangle \neq 0$ and $\langle \mathbf{1}, \mathbf{f}_\ell \rangle = 0, \forall \ell \in \{2, 3, \dots\}$.

This assumption allows us to consider only the minimal number of terms in the value function.

5. A NUMERICAL SIMULATION ON TRADE-OFF EFFECTS BETWEEN CONFORMISM AND INDIVIDUALISM

Suppose that we restrict $r_1 + r_2 = 1$, and $\delta := r_2/r_1$. Let the Assumption 16 be in force. Increasing r_1 penalizes individual deviations, while increasing r_2 penalizes deviations from the control mean field, so $\delta := r_2/r_1$ measures the conformism-individualism balance.

We consider scenarios for two graphons. In the first scenario, consider the Erdős-Renyi (ER) graph. Thus, its only eigenfunctions associated with non-zero eigenvalues is $\mathbf{f}_1(\alpha) = 1$. Although it does not form a basis of $L^2[0, 1]$, it will not invalidate the analysis in this work as $\mathbf{m} \in \text{span}\{\mathbf{f}_1\}$.

In the second scenario, suppose $\mathbf{g}_1$ has the following form,

$$\mathbf{g}_1(\alpha) = \sqrt{0.995} + 0.1 \sin(2\pi\alpha). \quad (56)$$

Two scenarios share the same parameters be as follows,

$$\begin{aligned}
b &= 0.2, \quad \sigma = 1, \quad \rho = 1, \quad \lambda_1 = 0.1, \quad \nu = 1, \quad m = 1, \\
\Gamma &= -\frac{\rho}{2b^2} + \sqrt{\frac{\rho^2}{4b^4} + \frac{1}{b^2}} \approx 0.8114. \quad (57)
\end{aligned}$$

For the first scenario, numerical simulation shows that there exists an optimal conformism-individualism ratio $\delta_\star$ that minimizes the value function uniformly in α . In contrast, in the second scenario, the optimal ratio $\delta_\star$ varies with the network location α , revealing a non-uniform trade-off that depends on the local network structure.

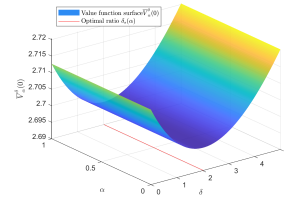


Fig. 1. Value function $\bar{V}_\alpha^\delta(0)$ in the ER graph case

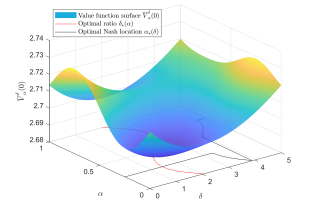


Fig. 2. Value function $\bar{V}_\alpha^\delta(0)$ in the second case

6. CONCLUSIONS

This work analyze the existence of the NE in a LQG-GMFG with control mean field cost, ensuring consistency of both the state and control mean fields. The results provide methods and insights for extending the analysis to problems with hard constraints parameterized by the control mean field. Huang (2026)

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